



Smriti Complete Codebase Architecture Dossier

End-to-End Technical System Design, Client/Server Component Hierarchy & Microservice Telemetry Pipeline

HACKATHON / MINISTRY

SIH26003 • Ministry of DoNER

FRONTEND TECH

React 19.2 + Vite 8.2 + PWA

BACKEND & DATABASE

Node.js + Express + MongoDB Atlas

AI & SPEECH ENGINES

Gemini AI + Bhashini TTS + Render ML

1. High-Level System Architecture & 3-Tier Delivery Model

Smriti is architected as an offline-first, highly resilient 3-Tier cognitive healthcare ecosystem designed to operate reliably in low-bandwidth, rural North-Eastern India (NER). The system seamlessly unites senior patient interfaces, clinician command centers, automated background cron dispatchers, and live cloud machine learning microservices.

THE SMRITI CODEBASE ARCHITECTURE

1. FRONTEND CLIENT LAYER	
React 19.2 SPA • TailwindCSS v4 • Vite 8.2 PWA • Workbox Service Worker • IndexedDB Sync Queue	
SENIOR PATIENT PORTAL (/patient/*) - Large 56px Touch Targets • Tactile 4-Digit PIN - Northeast State Theming (Assam, Mizo, etc.) - Horizon 10-Routine Timeline Spotlight 5 Culturally Attuned Cognitive Games Suite - Family Reminiscence Photo Vault (DPDP-Audited)	CLINICAL CAREGIVER HUB (/caregiver/*) - Deep Forest Green SaaS Design (#0B251E) - Real-time MongoDB Patient Roster Management - Longitudinal 7-Day Performance Telemetry - Live Red Flag Overdue Medication Alerts 3-Item Truncated History + Expand All Toggles
2. MESSAGING & VOICE AI PIPELINE	
DIGITAL INDIA BHASHINI TTS SYNTHESIS - Base64 WAV Streaming (/synthesize_speech) - Native Assamese, Bodo, Hindi Voice Generation - Graceful Fallback to Web Speech API	GOOGLE GEMINI MULTIMODAL COMPANION - Grounded Patient Context (Vitals, Meds) - Voice Audio Input + Text-to-Speech Streaming - Floating Patient Companion Widget
META WHATSAPP BUSINESS CLOUD API ENGINE - Two-way Conversational Adherence Bot - wa.me/15556680031 Automated Notification Bridge	AUTOMATED NODE-CRON SCHEDULER ENGINE - Continuous Prescription Timeline Polling - Caregiver Alert Dispatch & Escalation Rules
3. BACKEND GATEWAY & PERSISTENCE LAYER	
Node.js + Express.js 5.x REST API • Scoped JWT & WebAuthn FIDO2 Auth • MongoDB Atlas Cloud ODM	
LIVE PYTHON ML MICROSERVICE (RENDER CLOUD) - Endpoint: dementia-ai-engine.onrender.com - POST /get_next_difficulty (Adaptive Algorithm) - POST /calculate_health_score (Clinical Score)	MONGODB ATLAS CLOUD COLLECTIONS - Patients, Reminders, GameSessions, Caregivers, MemoryBankPhotos, ChatMessages - DPDP 2023 Consent Logs & 30-Day Erasure Queue

TIER 1: CORE CLIENT

Offline-First PWA

100% playable cognitive games, local medication check-offs, and IndexedDB sync queues operating during full network loss.

TIER 2: MESSAGING

WhatsApp Bot Gateway

Zero-app-learning curve routine alerts dispatched directly to rural caregiver and patient WhatsApp accounts.

TIER 3: AI & TELEMETRY

Bhashini & Live ML

Real-time Base64 WAV speech synthesis for Assamese and dynamic CST difficulty scaling powered by Render FastAPI.

2. Frontend Codebase Architecture (React 19 + Vite)

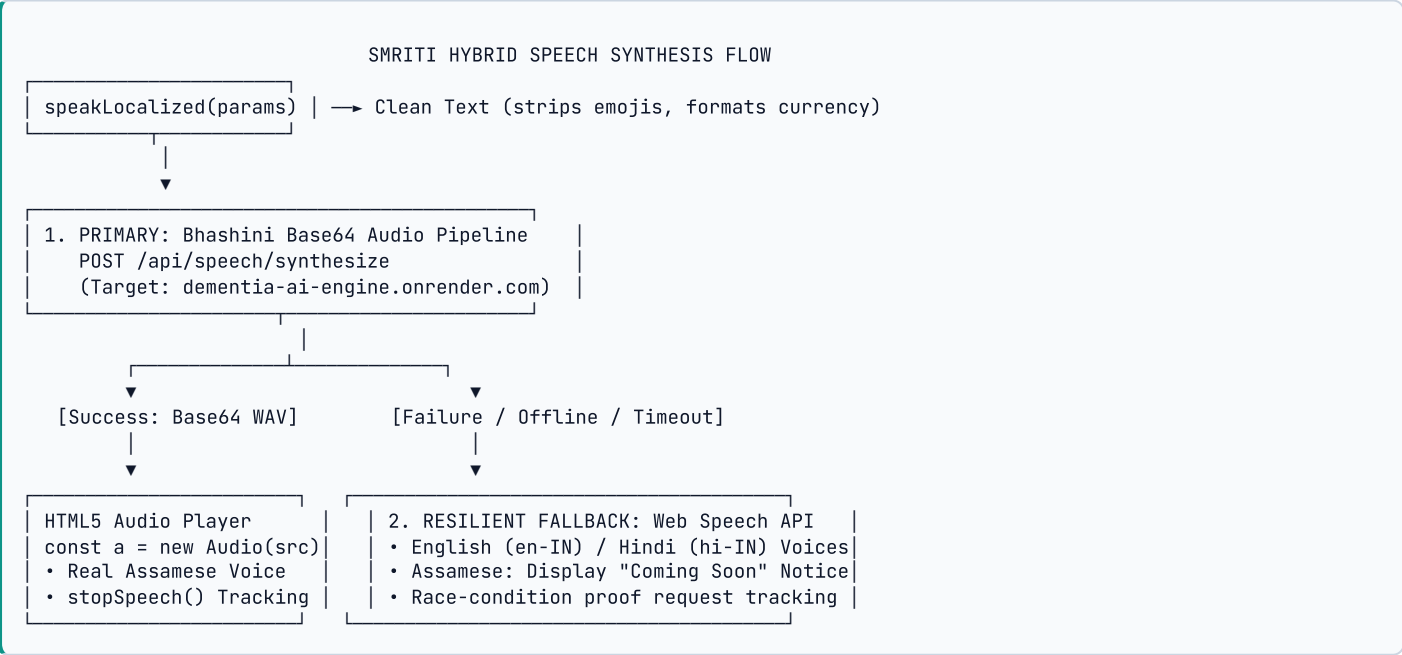
The frontend source is organized under frontend/src/ into strictly modularized components, pages, custom hooks, utilities, and services. It enforces clean separation of concerns between patient-facing accessibility and clinical data visualization.

Directory & Component File Map

Module Directory	Key Source Files	Architectural Role & Implementation Detail
src/pages/patient/	PatientDashboard.jsx PatientReminders.jsx PatientFamily.jsx PatientProfile.jsx PatientAllGames.jsx	Senior patient interface. Implements large 56px touch targets, time-aware horizon greetings, North-Eastern state horizon theming (Assam, Meghalaya, etc.), and spotlight routine progression.
src/pages/patient/games/	MarketDayBasket.jsx DailyRoutineSequencer.jsx FacesFamilyRecall.jsx SoundRhythmMatch.jsx OddOneOut.jsx	5 Culturally Attuned Cognitive Games Suite mapped to clinical CST domains: Episodic Memory, Executive Sequencing, Facial Recall, Auditory Discrimination, and Semantic Categorization.
src/pages/caregiver/	CaregiverDashboard.jsx CaregiverPatientDetail.jsx CaregiverCognitiveGames.jsx CaregiverNotifications.jsx CaregiverProfile.jsx	Clinical SaaS portal with Deep Forest Green styling (#0B251E). Features patient roster CRUD, AI cognitive scoring, 3-item default history limits with "Expand All" toggles, and Red Flag alert resolution.
src/utils/	speechUtils.js streakUtils.js webauthn.js	Speech synthesis engine with Bhashini Base64 HTML5 Audio player + Web Speech fallback, dynamic patient streak algorithms, and FIDO2 biometric authentication.
src/services/	api.js idbSync.js	Centralized Axios/Fetch API client with timeout controllers, direct fallback bridges, and IndexedDB sync queues for 100% offline data integrity.

Audio & Speech Synthesis Architecture (speechUtils.js)

Smriti implements a hybrid dual-engine speech pipeline that eliminates reliance on missing operating system voice packs for regional Indian languages:



⚙️ 3. Backend Architecture & REST API Gateway (Node.js + Express)

The backend server is structured under backend/ as a high-throughput, microservice-connected REST gateway using Express.js 5.x and Mongoose 9.x ODM on MongoDB Atlas Cloud.

REST API Route Catalog

Method	Endpoint Route	Controller / Service File	Payload & Clinical Functionality
POST	/api/speech/synthesize	translationRoutes.js	Synthesizes text into Base64 WAV audio via Bhashini TTS engine for regional dialects.
POST	/api/speech/translate	translationRoutes.js	Translates reminder and system prompts into Assamese (as), Bengali (bn), Hindi.
GET	/api/patients?batch=true	patientRoutes.js	Retrieves authenticated caregiver's patient roster with batch-calculated reminder compliance.
POST	/api/patients/login	patientRoutes.js	Validates senior tactile 4-digit PIN against bcrypt hash or biometric FIDO2 credential.
PATCH	/api/reminders/:id	reminderRoutes.js	Toggles routine completion, updates adherence timeline, and cancels active alert escalation.
GET	/api/reminders/alerts	reminderRoutes.js	Fetches real-time active Red Flag alerts (>3 hours unacknowledged critical prescriptions).
POST	/api/game-sessions	gameRoutes.js	Ingests millisecond round telemetry, score, mistakes, and proxies adaptive difficulty to ML engine.
GET	/api/game-sessions/ml-health-score/:id	gameRoutes.js	Fetches longitudinal AI Cognitive Health Score (0-100) and clinical category from Render ML.
POST	/api/whatsapp	whatsappWebhook.js	Meta WhatsApp Cloud API Webhook receiver; synchronizes patient chat responses with MongoDB.

Database Schema Models (backend/models/)

👤 Patient Schema (patient.js)

Stores patient identity, 4-digit bcrypt PIN hash, WebAuthn credentials, emergency contacts, primary doctor vitals, preferred NER state/language, and 10 daily routine reminders sub-documents.

👨 Caregiver Schema (Caregiver.js)

Clinician profile, bcrypt password hash, Google OAuth sub-ID, notification preferences (WhatsApp, SMS, Email), clinic affiliation, and associated patient reference arrays.

🎮 GameSession Schema (GameSession.js)

Records CST game type, timestamp, score, difficulty level, duration, and granular roundDetails (accuracy %, attempts, correct count, time taken in seconds).

🕒 Reminder Schema (Reminder.js)

Individual routine slot, scheduled time, period (Morning/Night), type (medicine, hydration, meal, game), status, acknowledged timestamp, and alert escalation flag.

🧠 4. Live Machine Learning & Gemini Multimodal AI Integration

Smriti bridges live AI services across two specialized cloud layers: Google Gemini AI SDK for conversational companionship, and an independent Python FastAPI microservice on Render Cloud for deterministic cognitive metrics.

MICROSERVICE: RENDER CLOUD

Python FastAPI Cognitive ML Engine

Host: <https://dementia-ai-engine.onrender.com>

- POST /synthesize_speech: Digital India Bhashini TTS Base64 audio stream.
- POST /get_next_difficulty: Calculates next starting level based on reaction time and mistake curves.
- POST /calculate_health_score: Evaluates 7-day CST performance into a 0-100 Clinical Index.
- Resilience:** Node backend implements a 6-second timeout with local rule-based heuristic fallback.

CONVERSATIONAL AI

Google Gemini Multimodal Companion

SDK: @google/genai

- Grounded in active patient clinical context (medication schedules, doctor contact, family names).
- Processes natural speech voice input and provides empathetic, dementia-safe replies.
- Supports multilingual text and audio synthesis across Assamese, Hindi, and English.

🔒 5. Security, Hardware Biometrics & DPDP Act 2023 Compliance

Hardware WebAuthn FIDO2

Public-key cryptographic authentication via on-device fingerprint/Face sensors. Raw biometric data never leaves the senior's device.

DPDP Act 2023 Consent

Explicit consent audit checkboxes before Memory Bank photo uploads, explicit purpose limitation, and transparent privacy terms.

30-Day Right to Erasure

Caregivers can trigger account deletion, executing an immediate database cascade and 30-day encrypted purge audit trail.

🔄 6. End-to-End Cognitive Telemetry Sequence Flow

